

“Climate scenarios and integrated assessment models”

Handout: DICE model — Sorbonne Université, M2 Actuariat

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Model Equations (DICE-style)

Time, Technology, Population

$$t(i) = t_0 + \Delta i, \quad (1)$$

$$A(i) = \frac{A(i-1)}{1 - g_A \exp(-\delta_A \Delta i)}, \quad (2)$$

$$L(i) = L(i-1) \left(\frac{L_\infty}{L(i-1)} \right)^{\ell_g}. \quad (3)$$

Carbon Intensity & Abatement Cost Share

$$\sigma(i) = \sigma(i-1) \exp\left(-g_\sigma (1 - \delta_\sigma)^{\Delta(i-1)} \Delta\right), \quad (4)$$

$$\theta_1(i) = \frac{p_b}{1000 \theta_2} (1 - \delta_{pb})^{i-1} \sigma(i). \quad (5)$$

Output, Damages/Abatement, Net Output

$$Y(i) = A(i) K(i-1)^\alpha \left(\frac{L(i)}{1000} \right)^{1-\alpha}, \quad (6)$$

$$Q(i) = \Delta \frac{(1 - \theta_1(i) \mu(i))^{\theta_2}}{1 + a_2 T_{AT}(i)^{a_3}} Y(i). \quad (7)$$

Consumption, Investment, Capital Accumulation

$$C(i) = (1 - s(i)) Q(i), \quad (8)$$

$$K(i) = (1 - \delta_K)^\Delta K(i-1) + s(i) Q(i). \quad (9)$$

Emissions & Radiative Forcing

$$E(i) = \Delta \left[\sigma(i) (1 - \mu(i)) Y(i) + E_{\text{Land}}(i) \right], \quad (10)$$

$$E_{\text{Land}}(i) = E_{L0} (1 - \delta_{EL})^i, \quad (11)$$

$$F_{EX}(i) = f_0 + \min \left(\frac{(f_1 - f_0)i}{t_f}, f_1 - f_0 \right), \quad (12)$$

$$F(i) = F_{2 \times \text{CO}_2} \log_2 \left(\frac{M_{AT}(i-1)}{M_{AT0}} \right) + F_{EX}(i). \quad (13)$$

Carbon Cycle (Three-Box)

$$M_{AT}(i) = (1 - \Delta b_{12}) M_{AT}(i-1) + \Delta b_{12} \frac{M_{AT0}}{M_{UP0}} M_{UP}(i-1) + \xi E(i), \quad (14)$$

$$M_{UP}(i) = \Delta b_{12} M_{AT}(i-1) + \left(1 - \Delta b_{12} \frac{M_{AT0}}{M_{UP0}} - \Delta b_{23} \right) M_{UP}(i-1) + \Delta b_{23} \frac{M_{UP0}}{M_{LO0}} M_{LO}(i-1), \quad (15)$$

$$M_{LO}(i) = \Delta b_{23} M_{UP}(i-1) + \left(1 - \Delta b_{23} \frac{M_{UP0}}{M_{LO0}} \right) M_{LO}(i-1). \quad (16)$$

Temperature Dynamics (Two-Box)

$$T_{AT}(i) = T_{AT}(i-1) + \Delta c_1 F(i) - \Delta c_1 \frac{F_{2 \times \text{CO}_2}}{T_{2 \times \text{CO}_2}} T_{AT}(i-1) - \Delta c_1 c_3 [T_{AT}(i-1) - T_{LO}(i-1)], \quad (17)$$

$$T_{LO}(i) = T_{LO}(i-1) + \Delta c_4 [T_{AT}(i-1) - T_{LO}(i-1)]. \quad (18)$$

Planner's Objective (CRRA over consumption per capita)

$$c_{pc}(i) = 1000 \frac{C(i)}{L(i)}, \quad (19)$$

$$u(i) = \begin{cases} L(i) \frac{c_{pc}(i)^{1-\gamma} - 1}{1-\gamma}, & \gamma \neq 1, \\ L(i) \ln c_{pc}(i), & \gamma = 1, \end{cases} \quad (20)$$

$$W = \sum_{i \in \mathcal{I}} \beta(i) u(i), \quad \beta(i) = \left(\frac{1}{1+\rho} \right)^{\Delta(i-i_0)}. \quad (21)$$

Key Symbols

- Δ time step; ρ pure rate of time preference; γ CRRA.
- A TFP; K capital; L population; α capital share.
- σ carbon intensity; μ abatement rate; θ_1, θ_2 abatement cost params.

- a_2, a_3 damage function params; T_{AT}, T_{LO} temperatures (atmosphere, deep ocean).
- E emissions; E_{Land} land-use emissions; F forcing; F_{EX} exogenous forcing.
- Carbon boxes: M_{AT}, M_{UP}, M_{LO} with preindustrial baselines $M_{AT0}, M_{UP0}, M_{LO0}$.
- Flows b_{12}, b_{23} ; conversion ξ ; climate response c_1, c_3, c_4 ; sensitivities $F_{2 \times CO_2}, T_{2 \times CO_2}$.

Table 1: Calibration of Model Parameters

Symbol	Value	Unit	Description
<i>Time grid</i>			
Δ	5	years	Time step
t_0	2015	year	Start year (fixed)
t_T	2100	year	End year of simulations
n_T	–	periods	Number of time steps
<i>Social preferences and production</i>			
ρ	0.015	1/yr	Pure rate of time preference
γ	1.45	–	CRRA coefficient
δ_K	0.10	1/yr	Capital depreciation rate
α	0.30	–	Capital share in production
<i>Technology</i>			
A_0	5.115	–	Initial TFP level
g_A	0.076	1/yr	Initial TFP growth rate
δ_A	0.005	1/yr	TFP growth decline rate
K_0	220.0	trillions USD	Initial capital stock
<i>Population</i>			
ℓ_g	0.134	1/yr	Population growth rate parameter
L_∞	10500	millions	Asymptotic population
L_0	7403	millions	Initial population
<i>Decoupling</i>			
g_σ	0.0152	1/yr	Initial carbon intensity decline
δ_σ	0.00	1/yr	Decline decay rate
σ_0	0.3503	GtCO ₂ /trillion USD	Initial carbon intensity
<i>Abatement</i>			
p_b	550	USD/tCO ₂	Initial backstop price
θ_2	2.6	–	Abatement cost exponent
δ_{pb}	0.025	1/yr	Decline rate of backstop price
<i>Climate damages</i>			
a_2	0.00236	–	Damage coefficient
a_3	2	–	Damage exponent

Table 2: Calibration of Model Parameters (cont)

Symbol	Value	Unit	Description
<i>Climate system</i>			
$F_{2 \times CO_2}$	3.3613	W/m ²	Forcing from CO ₂ doubling
$T_{2 \times CO_2}$	3.1	°C	Equilibrium climate sensitivity
ξ	3/11	CO ₂ to Carbon	Conversion factor CO ₂ emissions → carbon
M_{AT}^0	851	GtC	Initial atmospheric carbon
M_{UP}^0	460	GtC	Initial upper-ocean carbon
M_{LO}^0	1740	GtC	Initial deep-ocean carbon
M_{AT}	588	GtC	Preindustrial atmospheric carbon
M_{UP}	360	GtC	Preindustrial upper-ocean carbon
M_{LO}	1720	GtC	Preindustrial deep-ocean carbon
b_{12}	0.024	1/yr	Carbon flow: atmosphere → upper ocean
b_{23}	0.0014	1/yr	Carbon flow: upper ocean → deep ocean
c_1	0.0201	°C/(W/m ²)	Climate response coefficient
c_3	0.0176	1/yr	Heat transfer: atmosphere → ocean
c_4	0.005	1/yr	Heat transfer: ocean → atmosphere
T_{AT}^0	0.85	°C	Initial atmospheric temperature anomaly
T_{LO}^0	0.007	°C	Initial deep-ocean temperature anomaly
<i>Exogenous forcing</i>			
f_1	1.00	W/m ²	Final exogenous forcing
f_0	0.50	W/m ²	Initial exogenous forcing
t_f	17.0	years	Transition period for exogenous forcing
<i>Land-use emissions</i>			
E_{L0}	2.6	GtCO ₂ /yr	Initial land-use emissions
δ_{EL}	0.115	1/yr	Decline rate of land-use emissions
<i>Numerical bounds</i>			
$s_{\min}$	0.0	–	Lower bound on saving rate
$s_{\max}$	1.0	–	Upper bound on saving rate